

IOC Lookup Tool

A small Python tool I built to check IPs, file hashes, domains, URLs, and email headers against AbuseIPDB and VirusTotal — instead of doing it manually one at a time.

Why I built this

I was working through a home SOC lab (Wazuh + Atomic Red Team + OpenVAS) and kept manually pasting IOCs into different websites to check them. It got old fast, especially when checking more than a couple at once. I looked into how SOAR platforms automate this and decided to build a scaled-down version myself — partly to save time, mostly to actually understand what's happening instead of just clicking buttons in a tool.

It started as a single-IP checker. Once that worked, I added hashes, domains, URLs, and email header analysis one at a time, testing each before moving to the next.

What it does

- Checks IPs against AbuseIPDB (abuse score, country, ISP, report categories)
- Checks hashes, domains, and URLs against VirusTotal (detection ratio, tags, WHOIS for domains)
- Parses raw email headers — pulls sender IP, From domain, SPF/DKIM/DMARC results, and flags Reply-To mismatches to catch phishing
- Auto-detects what type of IOC you gave it, so you don't have to specify
- Two ways to run it: a terminal version for quick checks, and an Excel version that saves results and color-codes them (red/orange/green) so you can scan a batch of results at a glance

How it works (short version)

You give it an IOC → it figures out what type it is → sends it to the right API → reads back the JSON response → applies some simple threshold logic (e.g. 5+ AV engines flagging something = malicious) → prints or writes the result.

Tech used

Python, requests, openpyxl, AbuseIPDB API, VirusTotal API v3

Screenshots

Results from real test runs, both the PowerShell and Excel versions, across all supported IOC types.

Why two versions?

I initially built the PowerShell version, but results only printed to the terminal — nothing was saved, and reviewing more than a few results at once was hard to track. That's when I had the idea to move the same logic into Excel, so results would persist and be easier to scan visually with color coding. Both versions are shown here to demonstrate that progression.

Batch check — mixed IPs, hashes, domains, and URLs in one run (Powershell):

```
[1/8] Checking: 185.220.181.45
IOC      : 185.220.181.45
Type     : IP
Verdict  : MALICIOUS
Score    : 99%
Country  : DE
ISP      : Network for Tor-Exit traffic
Report Categories : Hacking, SQL Injection, Brute-Force, Bad Web Bot, Web App Attack, SSH
Last Reported : 2026-08-02

[2/8] Checking: 45.155.205.233
IOC      : 45.155.205.233
Type     : IP
Verdict  : CLEAN
Score    : 0%
Country  : RU
ISP      : Cloud Technologies LLC trading as Cloud.ru
Report Categories : Brute-Force, SQL Injection, SSH, Hacking
Last Reported : 2026-07-31

[3/8] Checking: e688012f6a8b73d6d2e1778a0b02f
C:\Users\james\Documents\Python\IP_lookup_v5.py:66: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
return datetime.datetime.fromtimestamp(timestamp).strftime('%Y-%m-%d')
IOC      : e688012f6a8b73d6d2e1778a0b02f
Type     : Hash
Verdict  : MALICIOUS
Detections : 53/79 engines
First Seen : 2006-05-22
Last Seen  : 2026-08-02
Tags      : via-tor, direct-cpu-clock-access, idle, long-sleeps, powershell, detect-debug-environment, attachment, known-distributor

[4/8] Checking: da39a3ee5e6b4b0d3255bf95661890af480709
IOC      : da39a3ee5e6b4b0d3255bf95661890af480709
Type     : Hash
Verdict  : CLEAN
Detections : 8/79 engines
First Seen : 2006-09-18
Last Seen  : 2026-08-02
Tags      : software-collection, known-distributor, nsrl, runtime-modules, zero-filled, direct-cpu-clock-access, via-tor, trusted, attachment, legit

[5/8] Checking: cim-kolea.com
IOC      : cim-kolea.com
Type     : Domain
Verdict  : SUSPICIOUS
Score    : 3/91 engines
Detections : 3/91 engines
Last Seen  : 2026-08-02
Tags      : None
WHOIS Registrar : NAMECHEAP INC
Domain Created : 2019-10-22 17:18:16.00Z

[6/8] Checking: google.com
IOC      : google.com
Type     : Domain
Verdict  : CLEAN
Score    : 0%
Country  : AU
ISP      : APNIC and Cloudflare DNS Resolver project
Report Categories : Brute-Force, SSH, Hacking, SQL Injection
Last Reported : 2026-08-02

[7/8] Checking: http://tustafabrowsersploit.com/s/malware.html
IOC      : http://tustafabrowsersploit.com/s/malware.html
Type     : URL
Verdict  : MALICIOUS
Score    : 12/74 engines
Detections : 12/74 engines
First Seen : 2016-08-02
Last Seen  : 2026-08-02
Tags      : contains-tip

[8/8] Checking: https://www.wikipedia.org
IOC      : https://www.wikipedia.org
Type     : URL
Verdict  : CLEAN
Score    : 8/74 engines
Detections : 8/74 engines
First Seen : 2012-06-15
Last Seen  : 2026-08-01
Tags      : None

SUMMARY
185.220.181.45      --> MALICIOUS
45.155.205.233     --> CLEAN
e688012f6a8b73d6d2e1778a0b02f --> MALICIOUS
da39a3ee5e6b4b0d3255bf95661890af480709 --> CLEAN
cim-kolea.com      --> SUSPICIOUS
google.com         --> CLEAN
http://tustafabrowsersploit.com/s/malware.html --> MALICIOUS
https://www.wikipedia.org --> CLEAN
```

IP Sheet (Excel)

IOC	Verdict	Score	Country	ISP	Report Categories	Last Reported
8.8.8.8	CLEAN	0%	US	Google LLC	Hacking, SQL Injection, Brute-Force, Exploited Host, SSH	2026-08-02
1.1.1.1	CLEAN	0%	AU	APNIC and Cloudflare DNS Resolver project	Brute-Force, SSH, Hacking, SQL Injection	2026-08-02
1319551c1c819465f1a797f6ccac79d494aaaf46fa3eac73ae63ffdbfd8267	MALICIOUS	100%	DE	Artike10 e.V.	Web Spam, Hacking, SQL Injection, Brute-Force, Bad Web Bot, Exploited Host, Web App Attack, SSH	2026-08-02
185.220.101.1	MALICIOUS	99%	DE	Network for Tor-Exit traffic.	Hacking, SQL Injection, Brute-Force, Bad Web Bot, Web App Attack, SSH	2026-08-02
45.155.205.233	CLEAN	6%	RU	Cloud Technologies LLC trading as Cloud.ru	Brute-Force, SQL Injection, SSH, Hacking	2026-07-31
115.49.66.100	CLEAN	0%	CN	China Unicom Henan province network	None	Never
46.227.184.200	CLEAN	15%	KZ	Kar-Tel LLC	Hacking, SQL Injection	2026-07-29
218.56.204.114	SUSPICIOUS	33%	CN	China Unicom Shandong province network	Brute-Force, Hacking, IoT Targeted	2026-08-02

Hash Sheet (Excel)

IOC	Verdict	Detections	First Seen	Last Seen	Tags
44d88612fea8a8f36de82e1278abb02f	MALICIOUS	65/79 engines	2006-05-22	2026-08-02	idle, detect-debug-environment, attachment, direct-cpu-clock-access, long-sleeps, powershell, via-tor, known-distributor
3395856ce81b2b7382dee726027798b642f14140	MALICIOUS	65/79 engines	2006-05-22	2026-08-02	idle, detect-debug-environment, attachment, direct-cpu-clock-access, long-sleeps, powershell, via-tor, known-distributor
1319551c1c819465f1a797f6ccac79d494aaaf46fa3eac73ae63ffdbfd8267	MALICIOUS	62/75 engines	2006-05-23	2026-07-31	idle, detect-debug-environment, powershell, known-distributor, attachment, long-sleeps, via-tor
275a021bbf6b6489e54d4718997db9d1663fc695ec2fe2a2c4538aabf651fd0	ERROR	HTTP 400			
d41d8cd98f00b204e9800998ec8427e	CLEAN	0/75 engines	2006-09-18	2026-08-02	direct-cpu-clock-access, runtime-modules, attachment, software-collection, trusted, known-distributor, via-tor, zero-filled, legit, nsrl
da39a3ee5e6b4b0d3255bf95661890af480709	CLEAN	0/75 engines	2006-09-18	2026-08-02	direct-cpu-clock-access, runtime-modules, attachment, software-collection, trusted, known-distributor, via-tor, zero-filled, legit, nsrl
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649ab934ca495991b7852b855	CLEAN	0/75 engines	2006-09-18	2026-08-02	direct-cpu-clock-access, runtime-modules, attachment, software-collection, trusted, known-distributor, via-tor, zero-filled, legit, nsrl
5d41402abc4b2a76b9719d911017c592	CLEAN	0/74 engines	2007-02-11	2026-07-22	text, trusted, nsrl, known-distributor, attachment, long-sleeps, idle
2aa6c35c94cfb415db95f40809ce91ee846ed	CLEAN	0/74 engines	2009-03-15	2026-07-15	long-sleeps, trusted, legit, known-distributor, idle, text
2cf24dba5f0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824	CLEAN	0/74 engines	2007-02-11	2026-07-22	text, trusted, nsrl, known-distributor, attachment, long-sleeps, idle
7bf2b57f2a20576875c0f7238fb32cc	MALICIOUS	64/74 engines	2017-05-12	2026-07-27	peexe, checks-user-input, runtime-modules, via-tor, direct-cpu-clock-access
2ef7bde608ce5404e975f042f95f89f0c34e6f9	NOT FOUND				
7c211433f02071597741e6f5a8ea34789abbf43729ba69007ed81c17c04447	ERROR	HTTP 400			
b026324c6904b2a9cb488d6d1c81d1	CLEAN	0/75 engines	2009-02-26	2026-08-02	idle, trusted, text, known-distributor, via-tor, long-sleeps, legit, nsrl
0cc17599cf0f16a831c399e269772661	CLEAN	0/74 engines	2008-05-21	2026-07-21	text, legit, nsrl, attachment, known-distributor, via-tor

Domain Sheet (Excel)

IOC	Verdict	Detections	Last Seen	Tags	WHOIS Registrar	Domain Created
google.com	CLEAN	0/91 engines	2026-08-0	None	MarkMonitor Inc.	1997-09-15
microsoft.com	CLEAN	0/91 engines	2026-08-0	None	Unknown	Unknown
cloudflare.com	CLEAN	0/91 engines	2026-08-0	None	Cloudflare, Inc.	2009-02-17
github.com	CLEAN	0/91 engines	2026-08-0	None	MarkMonitor Inc.	2007-10-09
wikipedia.org	CLEAN	0/91 engines	2026-08-0	None	Unknown	Unknown
cim-kolea.com	SUSPICIOUS	3/91 engines	2026-08-0	None	NAMECHEAP INC	2019-10-22

URL Sheet (Excel)

IOC	Verdict	Detections	First Seen	Last Seen	Tags
http://google.com	CLEAN	0/92 engines	2010-06-14	2026-08-02	external-resources, multiple-redirects
http://testsafebrowsing.appspot.com/s/malware.html	MALICIOUS	11/92 engines	2016-10-02	2026-08-02	contains-zip
https://github.com	CLEAN	0/92 engines	2011-04-25	2026-08-02	external-resources
https://www.wikipedia.org	CLEAN	0/92 engines	2012-06-15	2026-08-01	None
http://example.com	CLEAN	0/92 engines	2010-10-16	2026-08-02	None

Email Headers Sheet (Excel)

Header Text	Sender IP	From Domain	Reply-To Domain	SPF	DKIM	DMARC	Reply-To Mismatch	Sender IP Verdict	From Domain Verdict	Overall Verdict
Received: from mail-relay.google.com (209.85.220.41)	209.85.220.41	notifications.google.com	notifications.google.com	PASS	PASS	PASS	NO	CLEAN	CLEAN	LOW RISK
Received: from mail-fraud.suspicious-host.ru (203.0.113.77)	203.0.113.77	microsoft-security.com	fake-support-team.net	FAIL	FAIL	FAIL	YES	CLEAN	CLEAN	HIGH CONFIDENCE PHISHING
